BBQ SPECIAL

This VSAQ Section contains a few 1 Mark Qs (Q7 to Q 12)

VERY EASY & CUTE SECTION WITH MANY MINI ANSWERS

VSAQ SECTION-B

VSAQ CHAPTERS

10 Q X 2 M= 20 MARKS

★★	1.	RELATIONS & FUNCTIONS (RF) (2 Marks)	96 - 97
★★★	2.	MATRICES (MATS) (2 Marks)	98 - 102
★★★	3.	DETERMINANTS (DETS) (2 Marks)	103 - 104
★★★	4.	VECTOR ALGEBRA (VA) (2 Marks)	105 - 108
★★★	5.	3D GEOMETRY (3D) (2 Marks)	109 - 111
★★★	6.	PROBABILITY (PROB) (2 Marks)	112 - 114
★★★	7.	CONTINUITY & DIFFERENTIABILITY (CD) (2 Marks)	115 - 117
★★	8.	APPLICATION OF DERIVATIVES(AOD) (2 Marks)	118 - 119
★★★	9.	INTEGRATION (IG) (2 Marks)	120 - 124
★★★	10.	DIFFERENTIAL EQUATIONS (DE) (2 Marks)	125 - 127
•		STAR QUESTIONS PLUS (SQP)	128 - 129

Super Scoring Pages

★★★SSP★★★

Page 98 to 105, 110, 112, 115, 116, 118, 119, 120 to 124, 125, 127

Total SS Pages : 21

2 Marks

1 RELATIONS & FUNCTIONS

JUST 2 PAGES

153. Let A = {1, 2, 3}, B = {4, 5, 6, 7} and let f = {(1, 4), (2, 5), (3, 6)} be a function from A to B. Show that f is one-one.

Sol: Given that A = {1, 2, 3}, B = {4, 5, 6, 7}

f : A→B is defined as f = {(1, 4), (2, 5), (3, 6)}

∴ f(1) = 4, f(2) = 5, f(3) = 6

Thus, distinct elements of A have distinct images in B under f. ∴ f is one-one.

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154. Prove that the function f : R → R, given by f(x) = 2x, is one-one and onto.

Sol: f : R → R is given by f(x) = 2x. Let x_1, $x_2 \in R$ (domain)

(i) $f(x_1) = f(x_2) \Rightarrow 2x_1 = 2x_2 \Rightarrow x_1 = x_2$. ∴ f is one-one.

(ii) Let f(x) = y ⇒ y = 2x ⇒ $x = \frac{y}{2}$ exists ∀ y∈R (the codomain).

Now, $f(x) = 2x \Rightarrow f\left(\frac{y}{2}\right) = \cancel{2}\left(\frac{y}{\cancel{2}}\right) = y$. ∴ f is onto.

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155. Check the injectivity and surjectivity of the function f : N → N given by $f(x) = x^3$ **[BMP]**

Sol: f : N → N; $f(x) = x^3$

(i) $f(x_1) = f(x_2) \Rightarrow x_1^3 = x_2^3 \Rightarrow x_1 = x_2$ ∴ f is injective.

(ii) For 2∈N we can't find in x in N such that $f(x) = x^3 = 2$ ∴ f is not surjective

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156. Show that the function f : R → R given by $f(x) = x^3$ is injective.

Sol: f: R → R is given by $f(x) = x^3$. Let $x_1, x_2 \in R$ (domain)

Now $f(x_1) = f(x_2) \Rightarrow x_1^3 = x_2^3 \Rightarrow x_1 = x_2$. ∴ f is injective

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157. Check the injectivity and surjectivity of the function f : Z → Z given by $f(x) = x^2$

Sol: f : Z → Z; $f(x) = x^2$

(i) $f(x_1) = f(x_2) \Rightarrow x_1^2 = x_2^2 \Rightarrow x_1 = \pm x_2$ ∴ f is not injective.

(ii) For 2∈Z we can't find in x in Z such that $f(x) = x^2 = 2$ ∴ f is not surjective

158. **Show that the function $f: R \to R$, defined as $f(x) = x^2$, is neither one-one nor onto.**

Sol: (i) Given $f: R \to R$ is defined as $f(x) = x^2$. Take elements 1,–1 in the domain R.

Now $f(-1) = (-1)^2 = 1$; $f(1) = 1^2 = 1$

Thus, two different elements –1,1 of the domain have same image.

So, f is **not one-one.**

(ii) $f(x)=x^2 \Rightarrow$ Range of f contains all positive reals.

Moreover, Range does not contain any negative reals.

Given codomain is R. $\therefore$ Range $\neq$ Codomain.

So, f is **not onto.**

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159. **Define a reflexive relation**

Sol: **Reflexive Relation:** A relation R on a set A is reflexive if $(a,a) \in R$ for all $a \in A$.

Ex : The relation equality (=) on reals.

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160. **Define symmetric relation**

Sol: **Symmetric Relation:** A relation R on a set A is symmetric if $(a,b) \in R \Rightarrow (b,a) \in R$

Ex: The relation 'parallel of' lines ($||$) in a plane.

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161. **Define transitive relation**

Sol: **Transitive Relation:** A relation R on a set A is transitive

if $(a,b) \in R$ and $(b,c) \in R \Rightarrow (a,c) \in R$

Ex: The relation greater than (>) on reals.

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162. **Define an equivalence relation**

Sol: **Equivalence Relation:**

A relation R on a set A is called an equivalence relation

if R is reflexive, symmetric and transitive.

Ex: The relation equality (=) on reals.

2 MATRICES

MOST FAVOURITE CHAPTER TO ALL

163. If a matrix has 8 elements, what are the possible orders it can have?

Sol: If a matrix has 8 elements, then possible orders of the matrix have '8 as product'.
Hence, possible orders are $1 \times 8, 8 \times 1, 4 \times 2, 2 \times 4$

PQ **If a matrix has 24 elements, what are the possible orders it can have?**
What, if it has 13 elements? [Ans: (1×13) and (13×1)] **[For Answer ref. QR Code on P.No.126]**

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164. Find X, if $Y=\begin{bmatrix}3 & 2\\ 1 & 4\end{bmatrix}$ and $2X+Y=\begin{bmatrix}1 & 0\\ -3 & 2\end{bmatrix}$

Sol: $2X+Y=\begin{bmatrix}1 & 0\\ -3 & 2\end{bmatrix} \Rightarrow 2X+\begin{bmatrix}3 & 2\\ 1 & 4\end{bmatrix}=\begin{bmatrix}1 & 0\\ -3 & 2\end{bmatrix}$

$\Rightarrow 2X=\begin{bmatrix}1 & 0\\ -3 & 2\end{bmatrix}-\begin{bmatrix}3 & 2\\ 1 & 4\end{bmatrix}=\begin{bmatrix}-2 & -2\\ -4 & -2\end{bmatrix} \Rightarrow X=\frac{1}{2}\begin{bmatrix}-2 & -2\\ -4 & -2\end{bmatrix}=\begin{bmatrix}-1 & -1\\ -2 & -1\end{bmatrix}$.

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165. Find X and Y, if $X+Y=\begin{bmatrix}7 & 0\\ 2 & 5\end{bmatrix}$ and $X-Y=\begin{bmatrix}3 & 0\\ 0 & 3\end{bmatrix}$

Sol: Given $X+Y=\begin{bmatrix}7 & 0\\ 2 & 5\end{bmatrix}$.....(1) ; $X-Y=\begin{bmatrix}3 & 0\\ 0 & 3\end{bmatrix}$.....(2)

(1) + (2)$\Rightarrow$ $2X=\begin{bmatrix}7 & 0\\ 2 & 5\end{bmatrix}+\begin{bmatrix}3 & 0\\ 0 & 3\end{bmatrix}=\begin{bmatrix}10 & 0\\ 2 & 8\end{bmatrix}$

$\Rightarrow X=\frac{1}{2}\begin{bmatrix}10 & 0\\ 2 & 8\end{bmatrix}=\begin{bmatrix}5 & 0\\ 1 & 4\end{bmatrix} \Rightarrow X+Y=\begin{bmatrix}7 & 0\\ 2 & 5\end{bmatrix} \Rightarrow Y=\begin{bmatrix}7 & 0\\ 2 & 5\end{bmatrix}-\begin{bmatrix}5 & 0\\ 1 & 4\end{bmatrix} \Rightarrow Y=\begin{bmatrix}2 & 0\\ 1 & 1\end{bmatrix}$

PQ **Find X and Y, if $X+Y=\begin{bmatrix}5 & 2\\ 0 & 9\end{bmatrix}$ and $X-Y=\begin{bmatrix}3 & 6\\ 0 & -1\end{bmatrix}$** $\left[A: X=\begin{bmatrix}4 & 4\\ 0 & 4\end{bmatrix}, Y=\begin{bmatrix}1 & -2\\ 0 & 5\end{bmatrix}\right]$
[For Answer ref. QR Code on P.No.128]

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166. Find x and y, if $2\begin{bmatrix}1 & 3\\ 0 & x\end{bmatrix}+\begin{bmatrix}y & 0\\ 1 & 2\end{bmatrix}=\begin{bmatrix}5 & 6\\ 1 & 8\end{bmatrix}$

ol: Given $2\begin{bmatrix}1 & 3\\ 0 & x\end{bmatrix}+\begin{bmatrix}y & 0\\ 1 & 2\end{bmatrix}=\begin{bmatrix}5 & 6\\ 1 & 8\end{bmatrix}$

$\Rightarrow \begin{bmatrix}2 & 6\\ 0 & 2x\end{bmatrix}+\begin{bmatrix}y & 0\\ 1 & 2\end{bmatrix}=\begin{bmatrix}5 & 6\\ 1 & 8\end{bmatrix} \Rightarrow \begin{bmatrix}2+y & 6\\ 1 & 2x+2\end{bmatrix}=\begin{bmatrix}5 & 6\\ 1 & 8\end{bmatrix}$

Equating the corresponding elements, $2+y=5 \Rightarrow y=3$; $2x+2=8 \Rightarrow x=3$ $\therefore x=3, y=$

167. If $x\begin{bmatrix}2\\3\end{bmatrix}+y\begin{bmatrix}-1\\1\end{bmatrix}=\begin{bmatrix}10\\5\end{bmatrix}$ find the values of x and y.

Sol: $x\begin{bmatrix}2\\3\end{bmatrix}+y\begin{bmatrix}-1\\1\end{bmatrix}=\begin{bmatrix}10\\5\end{bmatrix}\Rightarrow\begin{bmatrix}2x\\3x\end{bmatrix}+\begin{bmatrix}-y\\y\end{bmatrix}=\begin{bmatrix}10\\5\end{bmatrix}\Rightarrow\begin{bmatrix}2x-y\\3x+y\end{bmatrix}=\begin{bmatrix}10\\5\end{bmatrix}$

Equating the corresponding elements $2x-y=10$.....(1); $3x+y=5$.....(2)

(1) + (2) $\Rightarrow 5x=15\Rightarrow x=3$

(2) $\Rightarrow 3x+y=5\Rightarrow y=5-3x \Rightarrow y=5-3(3)\Rightarrow y=5-9 \Rightarrow y=-4$ $\quad\therefore$ x=3, y= –4

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168. Find the values of x,y and z from $\begin{bmatrix}x+y+z\\x+z\\y+z\end{bmatrix}=\begin{bmatrix}9\\5\\7\end{bmatrix}$

Sol: Equating the corresponding elements, we get

$x+y+z=9$.....(1); $x+z=5$.....(2); $y+z=7$.....(3)

From (1) and (2), we have $y+5=9\Rightarrow y=4$

From (3), we have $4+z=7\Rightarrow z=3$

Now $x+z=5\Rightarrow x+3=5\Rightarrow x=2$. Thus, $x=2, y=4, z=3$

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169. Find the values of x,y and z from $\begin{bmatrix}x+y & 2\\5+z & xy\end{bmatrix}=\begin{bmatrix}6 & 2\\5 & 8\end{bmatrix}$ **[BMP]**

Sol: Given $\begin{bmatrix}x+y & 2\\5+z & xy\end{bmatrix}=\begin{bmatrix}6 & 2\\5 & 8\end{bmatrix}$

Equating the corresponding elements, we get $x+y=6$; $xy=8$; $5+z=5\Rightarrow z=0$

Now $(x-y)^2=(x+y)^2-4xy\Rightarrow(x-y)^2=6^2-4(8)=36-32=4$

$\Rightarrow(x-y)^2=4\Rightarrow(x-y)=\pm2\Rightarrow x-y=2$ or $x-y=-2$

When $x-y=2$ and $x+y=6$ we get $x=4, y=2$

When $x-y=-2$ and $x+y=6$ we get $x=2, y=4$

Thus, $x=4, y=2, z=0$ or $x=2, y=4, z=0$

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170. Compute $\begin{bmatrix}\cos^2x & \sin^2x\\\sin^2x & \cos^2x\end{bmatrix}+\begin{bmatrix}\sin^2x & \cos^2x\\\cos^2x & \sin^2x\end{bmatrix}$

Sol: $\begin{bmatrix}\cos^2x & \sin^2x\\\sin^2x & \cos^2x\end{bmatrix}+\begin{bmatrix}\sin^2x & \cos^2x\\\cos^2x & \sin^2x\end{bmatrix}=\begin{bmatrix}\cos^2x+\sin^2x & \sin^2x+\cos^2x\\\sin^2x+\cos^2x & \cos^2x+\sin^2x\end{bmatrix}=\begin{bmatrix}1 & 1\\1 & 1\end{bmatrix}$

171. Construct a 2×2 matrix, $A = [a_{ij}]$, whose elements are given by $a_{ij} = \frac{(i+j)^2}{2}$

Sol: A 2×2 matrix is given by $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$

Given that $a_{ij} = \frac{(i+j)^2}{2}$; $i, j = 1, 2$

$\therefore a_{11} = \frac{(1+1)^2}{2} = \frac{4}{2} = 2$; $a_{12} = \frac{(1+2)^2}{2} = \frac{9}{2}$;

$a_{21} = \frac{(2+1)^2}{2} = \frac{9}{2}$; $a_{22} = \frac{(2+2)^2}{2} = \frac{16}{2} = 8$

$\therefore A = \begin{bmatrix} 2 & 9/2 \\ 9/2 & 8 \end{bmatrix}$

PQ Construct a 2×2 matrix whose elements are $a_{ij} = \frac{i}{j}$ $\left[\text{Ans}: A = \begin{bmatrix} 1 & 1/2 \\ 2 & 1 \end{bmatrix}\right]$

PQ Construct a 3×2 matrix whose elements are $a_{ij} = \frac{1}{2}|i-3j|$ $\left[\text{Ans: } A = \begin{bmatrix} 1 & 5/2 \\ 1/2 & 2 \\ 0 & 3/2 \end{bmatrix}\right]$

[For Answers ref. QR Code on P.No.128]

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172. Let $A = \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix}, B = \begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix}$. Find AB

Sol: $AB = \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix}\begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix} = \begin{bmatrix} 2(1)+4(-2) & 2(3)+4(5) \\ 3(1)+2(-2) & 3(3)+2(5) \end{bmatrix} = \begin{bmatrix} 2-8 & 6+20 \\ 3-4 & 9+10 \end{bmatrix} = \begin{bmatrix} -6 & 26 \\ -1 & 19 \end{bmatrix}$

PQ Find AB, if $A = \begin{bmatrix} 0 & -1 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 5 \\ 0 & 0 \end{bmatrix}$ $\left[\text{Ans}: \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}\right]$

PQ Let $A = \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix}, B = \begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix}$. Find BA $\left[\text{Ans}: BA = \begin{bmatrix} 11 & 10 \\ 11 & 2 \end{bmatrix}\right]$

[For Answers ref. QR Code on P.No.128]

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173. Compute $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}[2 \quad 3 \quad 4]$

Sol: $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}[2 \quad 3 \quad 4] = \begin{bmatrix} 1(2) & 1(3) & 1(4) \\ 2(2) & 2(3) & 2(4) \\ 3(2) & 3(3) & 3(4) \end{bmatrix} = \begin{bmatrix} 2 & 3 & 4 \\ 4 & 6 & 8 \\ 6 & 9 & 12 \end{bmatrix}$

PQ Compute $\begin{bmatrix} 1 & -2 \\ 2 & 3 \end{bmatrix}\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$ $\left[\text{Ans}: \begin{bmatrix} -3 & -4 & 1 \\ 8 & 13 & 9 \end{bmatrix}\right]$

PQ Compute $\begin{bmatrix} 3 & -1 & 3 \\ -1 & 0 & 2 \end{bmatrix}\begin{bmatrix} 2 & -3 \\ 1 & 0 \\ 3 & 1 \end{bmatrix}$ $\left[\text{Ans}: \begin{bmatrix} 14 & -6 \\ 4 & 5 \end{bmatrix}\right]$

[For Answers ref. QR Code on P.No.1

174. Find AB, if $A=\begin{bmatrix}6 & 9\\ 2 & 3\end{bmatrix}$ and $B=\begin{bmatrix}2 & 6 & 0\\ 7 & 9 & 8\end{bmatrix}$

Sol:
$$AB=\begin{bmatrix}6 & 9\\ 2 & 3\end{bmatrix}\begin{bmatrix}2 & 6 & 0\\ 7 & 9 & 8\end{bmatrix}=\begin{bmatrix}6(2)+9(7) & 6(6)+9(9) & 6(0)+9(8)\\ 2(2)+3(7) & 2(6)+3(9) & 2(0)+3(8)\end{bmatrix}$$
$$=\begin{bmatrix}12+63 & 36+81 & 0+72\\ 4+21 & 12+27 & 0+24\end{bmatrix}=\begin{bmatrix}75 & 117 & 72\\ 25 & 39 & 24\end{bmatrix}$$

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175. If $A=\begin{bmatrix}\sin\alpha & \cos\alpha\\ -\cos\alpha & \sin\alpha\end{bmatrix}$ then verify that $A'A=I$

Sol: Given that $A=\begin{bmatrix}\sin\alpha & \cos\alpha\\ -\cos\alpha & \sin\alpha\end{bmatrix}$ $\quad\therefore A'=\begin{bmatrix}\sin\alpha & -\cos\alpha\\ \cos\alpha & \sin\alpha\end{bmatrix}$

Now, $A'A=\begin{bmatrix}\sin\alpha & -\cos\alpha\\ \cos\alpha & \sin\alpha\end{bmatrix}\begin{bmatrix}\sin\alpha & \cos\alpha\\ -\cos\alpha & \sin\alpha\end{bmatrix}$

$$=\begin{bmatrix}\sin\alpha\sin\alpha+(-\cos\alpha)(-\cos\alpha) & \sin\alpha\cos\alpha+(-\cos\alpha)\sin\alpha\\ \sin\alpha\cos\alpha+\sin\alpha(-\cos\alpha) & \cos\alpha\cos\alpha+\sin\alpha\sin\alpha\end{bmatrix}$$
$$=\begin{bmatrix}\cos^2\alpha+\sin^2\alpha & \sin\alpha\cos\alpha-\sin\alpha\cos\alpha\\ \sin\alpha\cos\alpha-\sin\alpha\cos\alpha & \cos^2\alpha+\sin^2\alpha\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}=I$$

Thus, $A'A=I$.

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176. Find x, if $\begin{bmatrix}x & -5 & -1\end{bmatrix}\begin{bmatrix}1 & 0 & 2\\ 0 & 2 & 1\\ 2 & 0 & 3\end{bmatrix}\begin{bmatrix}x\\ 4\\ 1\end{bmatrix}=O$

Sol: Given that $\begin{bmatrix}x & -5 & -1\end{bmatrix}\begin{bmatrix}1 & 0 & 2\\ 0 & 2 & 1\\ 2 & 0 & 3\end{bmatrix}\begin{bmatrix}x\\ 4\\ 1\end{bmatrix}=O$

$$\Rightarrow\begin{bmatrix}x+0-2 & 0-10+0 & 2x-5-3\end{bmatrix}\begin{bmatrix}x\\ 4\\ 1\end{bmatrix}=[0]$$

$$\Rightarrow\begin{bmatrix}x-2 & -10 & 2x-8\end{bmatrix}\begin{bmatrix}x\\ 4\\ 1\end{bmatrix}=[0]\Rightarrow[x(x-2)-40+2x-8]=[0]$$

$$\Rightarrow[x^2-2x-40+2x-8]=[0]\Rightarrow[x^2-48]=[0]\Rightarrow x^2-48=0$$

$$\Rightarrow x^2=48\Rightarrow x=\pm4\sqrt{3}$$

PQ **For what values of x :** $\begin{bmatrix}1 & 2 & 1\end{bmatrix}\begin{bmatrix}1 & 2 & 0\\ 2 & 0 & 1\\ 1 & 0 & 2\end{bmatrix}\begin{bmatrix}0\\ 2\\ x\end{bmatrix}=O$ [Ans: x= –1]

[For Answer ref. QR Code on P.No.128]

177. Show that the matrix $A=\begin{bmatrix}1 & -1 & 5\\ -1 & 2 & 1\\ 5 & 1 & 3\end{bmatrix}$ is a symmetric matrix.

Sol: Given that $A=\begin{bmatrix}1 & -1 & 5\\ -1 & 2 & 1\\ 5 & 1 & 3\end{bmatrix}$

Now $A'=\begin{bmatrix}1 & -1 & 5\\ -1 & 2 & 1\\ 5 & 1 & 3\end{bmatrix}=A$ Hence, A is a symmetric matrix.

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178. Show that the matrix $A=\begin{bmatrix}0 & 1 & -1\\ -1 & 0 & 1\\ 1 & -1 & 0\end{bmatrix}$ is a skew symmetric matrix.

Sol: $A=\begin{bmatrix}0 & 1 & -1\\ -1 & 0 & 1\\ 1 & -1 & 0\end{bmatrix}$

Now $A'=\begin{bmatrix}0 & -1 & 1\\ 1 & 0 & -1\\ -1 & 1 & 0\end{bmatrix}=-\begin{bmatrix}0 & 1 & -1\\ -1 & 0 & 1\\ 1 & -1 & 0\end{bmatrix}=-A$ Hence, A is a skew symmetric matrix.

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179. For the matrix $A=\begin{bmatrix}1 & 5\\ 6 & 7\end{bmatrix}$, verify that $(A+A')$ is a symmetric matrix

Sol: Given that $A=\begin{bmatrix}1 & 5\\ 6 & 7\end{bmatrix}\Rightarrow A'=\begin{bmatrix}1 & 6\\ 5 & 7\end{bmatrix}$

$$A+A'=\begin{bmatrix}1 & 5\\ 6 & 7\end{bmatrix}+\begin{bmatrix}1 & 6\\ 5 & 7\end{bmatrix}=\begin{bmatrix}2 & 11\\ 11 & 14\end{bmatrix}$$

Also $[(A+A')]'=\begin{bmatrix}2 & 11\\ 11 & 14\end{bmatrix}=(A+A')$ $\therefore (A+A')$ is a symmetric matrix

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180. Compute $\begin{bmatrix}a^2+b^2 & b^2+c^2\\ a^2+c^2 & a^2+b^2\end{bmatrix}+\begin{bmatrix}2ab & 2bc\\ -2ac & -2ab\end{bmatrix}$

Sol: $\begin{bmatrix}a^2+b^2 & b^2+c^2\\ a^2+c^2 & a^2+b^2\end{bmatrix}+\begin{bmatrix}2ab & 2bc\\ -2ac & -2ab\end{bmatrix}=\begin{bmatrix}a^2+b^2+2ab & b^2+c^2+2bc\\ a^2+c^2-2ac & a^2+b^2-2ab\end{bmatrix}=\begin{bmatrix}(a+b)^2 & (b+c)^2\\ (a-c)^2 & (a-b)^2\end{bmatrix}$

2 Marks

3 DETERMINANTS

CUTE 8 PROBLEMS IN JUST 2 PAGES

181. Evaluate $\begin{vmatrix} x & x+1 \\ x-1 & x \end{vmatrix}$

Sol: We have $\begin{vmatrix} x & x+1 \\ x-1 & x \end{vmatrix} = x(x)-(x+1)(x-1) = x^2-(x^2-1) = x^2-x^2+1 = 1$

PQ Evaluate $\begin{vmatrix} x^2-x+1 & x-1 \\ x+1 & x+1 \end{vmatrix}$ [Ans: x^3-x^2+2]

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182. Evaluate $\begin{vmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{vmatrix}$

Sol: $\begin{vmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{vmatrix} = (\cos\theta)(\cos\theta)-(-\sin\theta)(\sin\theta) = \cos^2\theta+\sin^2\theta = 1$

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183. Find the values of x for which $\begin{vmatrix} 3 & x \\ x & 1 \end{vmatrix} = \begin{vmatrix} 3 & 2 \\ 4 & 1 \end{vmatrix}$

Sol: We have $\begin{vmatrix} 3 & x \\ x & 1 \end{vmatrix} = \begin{vmatrix} 3 & 2 \\ 4 & 1 \end{vmatrix}$ $\left[\because \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad-bc\right]$

$\Rightarrow 3-x^2 = 3-8 \Rightarrow x^2 = 8 \Rightarrow x = \pm\sqrt{8} = \pm\sqrt{4\times2} = \pm 2\sqrt{2}$

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184. Find values of x, if $\begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} = \begin{vmatrix} x & 3 \\ 2x & 5 \end{vmatrix}$

Sol: Given that $\begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} = \begin{vmatrix} x & 3 \\ 2x & 5 \end{vmatrix}$

$\Rightarrow 2\times5-3\times4 = x\times5-3\times2x \Rightarrow 10-12 = 5x-6x \Rightarrow -2 = -x \Rightarrow x = 2$

PQ Find the values of x, if $\begin{vmatrix} 2 & 4 \\ 5 & 1 \end{vmatrix} = \begin{vmatrix} 2x & 4 \\ 6 & x \end{vmatrix}$ [Ans: $x = \pm\sqrt{3}$]

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185. If $A = \begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix}$, then show that $|2A| = 4|A|$

Sol: The given matrix is $A = \begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix}$

$\therefore 2A = 2\begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 8 & 4 \end{bmatrix}$

$\left[\because \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad-bc\right]$

L.H.S $= |2A| = \begin{vmatrix} 2 & 4 \\ 8 & 4 \end{vmatrix} = 2\times4 - 4\times8 = 8 - 32 = -24$

Now, $|A| = \begin{vmatrix} 1 & 2 \\ 4 & 2 \end{vmatrix} = 1\times2 - 2\times4 = 2 - 8 = -6$ $\quad \therefore$ RHS $= 4|A| = 4(-6) = -24$

$\therefore |2A| = 4|A|$

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186. Write Minors and Cofactors of the elements of $\begin{vmatrix} 2 & -4 \\ 0 & 3 \end{vmatrix}$

Sol: The given determinant is $\begin{vmatrix} 2 & -4 \\ 0 & 3 \end{vmatrix}$

Minor of the element a_{ij} is M_{ij}

M_{11} = Minor of the element $a_{11} = 3$; $\quad M_{12}$ = Minor of the element $a_{12} = 0$;

M_{21} = Minor of the element $a_{21} = -4$; $\quad M_{22}$ = Minor of the element $a_{22} = 2$;

Now, cofactor of a_{ij} is $A_{ij} = (-1)^{i+j} M_{ij}$

$A_{11} = (-1)^{1+1}(3) = 3$; $\quad A_{12} = (-1)^{1+2}(0) = 0$

$A_{21} = (-1)^{2+1}(-4) = 4$; $\quad A_{22} = (-1)^{2+2}(2) = 2$

PQ **Write Minors and Cofactors of the elements of** $\begin{vmatrix} a & c \\ b & d \end{vmatrix}$

[Ans: Minors: d,b,c,a; Cofactors: d,–b,–c,a]

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187. Find adjA for $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$ **[BMP]**

Sol: We have $A_{11} = 4, A_{12} = -1, A_{21} = -3, A_{22} = 2$ $\quad \therefore$ adjA $= \begin{bmatrix} A_{11} & A_{21} \\ A_{12} & A_{22} \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ -1 & 2 \end{bmatrix}$

PQ **Find the adjoint of the matrix** $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $\quad \left[\text{Ans: adjA} = \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}\right]$

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188. Find the inverse of the matrix $\begin{bmatrix} 2 & -2 \\ 4 & 3 \end{bmatrix}$

Sol: Let $A = \begin{bmatrix} 2 & -2 \\ 4 & 3 \end{bmatrix}$. Then, $|A| = (2\times3) - (-2\times4) = 6 - (-8) = 14$

Now, $A_{11} = 3$; $\quad A_{12} = -4$ $\quad A_{21} = 2$; $\quad A_{22} = 2$

Hence, adjA $= \begin{bmatrix} 3 & 2 \\ -4 & 2 \end{bmatrix}$ $\quad \therefore A^{-1} = \frac{1}{|A|}\text{adjA} = \frac{1}{14}\begin{bmatrix} 3 & 2 \\ -4 & 2 \end{bmatrix}$

PQ **Find the inverse of the matrix** $\begin{bmatrix} -1 & 5 \\ -3 & 2 \end{bmatrix}$ $\quad \left[\therefore A^{-1} = \frac{1}{13}\begin{bmatrix} 2 & -5 \\ 3 & -1 \end{bmatrix}\right]$

2 Marks

4 VECTOR ALGEBRA

PHYSICS Qs IN MATHS

189. **Classify the following as scalar and vector quantities i)force ii) distance [BMP]**

Sol: i) Force-vector ii) Distance-scalar

PQ **Classify the following as scalar and vector quantities.**

i) **work done ii) displacement** [Ans: i) Work done - scalar. ii) displacement - vector]

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190. **Compute the magnitude of the vector $\vec{a}=\hat{i}+\hat{j}+\hat{k}$** **[BMP]**

Sol: Given that $\vec{a}=\hat{i}+\hat{j}+\hat{k}$ $\therefore |\vec{a}| = \sqrt{x^2+y^2+z^2} = \sqrt{1+1+1} = \sqrt{3}$

PQ Compute the magnitude of the vector $\vec{b}=2\hat{i}-7\hat{j}-3\hat{k}$

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191. **Find the sum of the vectors $\vec{a}=\hat{i}-2\hat{j}+\hat{k}$, $\vec{b}=-2\hat{i}+4\hat{j}+5\hat{k}$ and $\vec{c}=\hat{i}-6\hat{j}-7\hat{k}$.**

Sol: Given vectors $\vec{a}=\hat{i}-2\hat{j}+\hat{k}$, $\vec{b}=-2\hat{i}+4\hat{j}+5\hat{k}$ and $\vec{c}=\hat{i}-6\hat{j}-7\hat{k}$.

$\therefore \vec{a}+\vec{b}+\vec{c} = (1-2+1)\hat{i}+(-2+4-6)\hat{j}+(1+5-7)\hat{k} = 0\hat{i}-4\hat{j}-\hat{k} = -4\hat{j}-\hat{k}$

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192. **Find unit vector in the direction of vector $\vec{a}=2\hat{i}+3\hat{j}+\hat{k}$**

Sol: Given $\vec{a}=2\hat{i}+3\hat{j}+\hat{k} \Rightarrow |\vec{a}| = \sqrt{2^2+3^2+1^2} = \sqrt{14}$

$\therefore$ Unit vector $\hat{a} = \frac{1}{|\vec{a}|}\vec{a} = \frac{1}{\sqrt{14}}(2\hat{i}+3\hat{j}+\hat{k}) = \frac{2}{\sqrt{14}}\hat{i}+\frac{3}{\sqrt{14}}\hat{j}+\frac{1}{\sqrt{14}}\hat{k}$

PQ **Find the unit vector in the direction of the vector $\vec{a}=\hat{i}+\hat{j}+2\hat{k}$.** $\left[A: \frac{1}{\sqrt{6}}\hat{i}+\frac{1}{\sqrt{6}}\hat{j}+\frac{2}{\sqrt{6}}\hat{k}\right]$.

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193. **Find the unit vector in the direction of the sum of the vectors, $\vec{a}=2\hat{i}+2\hat{j}-5\hat{k}$ and $\vec{b}=2\hat{i}+\hat{j}+3\hat{k}$.**

Sol: Sum of the given vectors is $\vec{a}+\vec{b}\,(=\vec{c}, \text{say}) = 4\hat{i}+3\hat{j}-2\hat{k}$ and $|\vec{c}| = \sqrt{4^2+3^2+(-2)^2} = \sqrt{29}$

Thus, the required unit vector is $\hat{c} = \frac{1}{|\vec{c}|}\vec{c} = \frac{1}{\sqrt{29}}(4\hat{i}+3\hat{j}-2\hat{k}) = \frac{4}{\sqrt{29}}\hat{i}+\frac{3}{\sqrt{29}}\hat{j}-\frac{2}{\sqrt{29}}\hat{k}$

PQ **For given vectors, $\vec{a}=2\hat{i}-\hat{j}+2\hat{k}$ and $\vec{b}=-\hat{i}+\hat{j}-\hat{k}$. Find the unit vector in the direction of $\vec{a}+\vec{b}$.** $\left[\text{Ans}: \frac{1}{\sqrt{2}}\hat{i}+\frac{1}{\sqrt{2}}\hat{k}\right]$

194. **Find a vector in the direction of vector $5\hat{i}-\hat{j}+2\hat{k}$ which has magnitude 8 units.**

Sol: Let $\vec{a}=5\hat{i}-\hat{j}+2\hat{k}$. $\therefore$ A vector in the direction of vector $\vec{a}$ which has magnitude 8 units

$$=8\hat{a}=8\frac{\vec{a}}{|\vec{a}|}=\frac{8(5\hat{i}-\hat{j}+2\hat{k})}{\sqrt{25+1+4}}=\frac{8}{\sqrt{30}}(5\hat{i}-\hat{j}+2\hat{k})=\frac{40}{\sqrt{30}}\hat{i}-\frac{8}{\sqrt{30}}\hat{j}+\frac{16}{\sqrt{30}}\hat{k}$$

PQ **Find a vector in the direction of vector $\vec{a}=\hat{i}-2\hat{j}$ that has magnitude 7 units.**

$$\left[\text{Ans}: \frac{7}{\sqrt{5}}\hat{i}-\frac{14}{\sqrt{5}}\hat{j}\right]$$

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195. **Find the value of x for which $x(\hat{i}+\hat{j}+\hat{k})$ is a unit vector.**

Sol: If $x(\hat{i}+\hat{j}+\hat{k})$ is a unit vector then

$$\left|x(\hat{i}+\hat{j}+\hat{k})\right|=1\Rightarrow\sqrt{x^2+x^2+x^2}=1\Rightarrow\sqrt{3x^2}=1\Rightarrow\sqrt{3}x=1\Rightarrow x=\pm\frac{1}{\sqrt{3}}$$

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196. **Find $|\vec{x}|$, if for a unit vector $\vec{a}$, $(\vec{x}-\vec{a}).(\vec{x}+\vec{a})=12$**

Sol: Given that $\vec{a}$ is a unit vector $\Rightarrow|\vec{a}|=1$ .....(i)

Also $(\vec{x}-\vec{a}).(\vec{x}+\vec{a})=12\Rightarrow\vec{x}.\vec{x}+\cancel{\vec{x}.\vec{a}}-\cancel{\vec{a}.\vec{x}}-\vec{a}.\vec{a}=12\Rightarrow|\vec{x}|^2-|\vec{a}|^2=12$

Putting $|\vec{a}|=1$ from (i), $|\vec{x}|^2-1=12\Rightarrow|\vec{x}|^2=13\Rightarrow|\vec{x}|=\sqrt{13}$.

PQ **If $\vec{a}$ is a unit vector and $(\vec{x}-\vec{a})\cdot(\vec{x}+\vec{a})=8$, then find $|\vec{x}|$.** **[Ans: $|\vec{x}|=3$]**

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197. **Show that the vectors $2\hat{i}-3\hat{j}+4\hat{k}$ and $-4\hat{i}+6\hat{j}-8\hat{k}$ are collinear.**

Sol: Let $\vec{a}=2\hat{i}-3\hat{j}+4\hat{k}$ and $\vec{b}=-4\hat{i}+6\hat{j}-8\hat{k}=-2(2\hat{i}-3\hat{j}+4\hat{k})=-2\vec{a}$

Here, $\vec{b}=-2\vec{a}=m\vec{a}$ where $m=-2<0$

$\therefore$ Vectors $\vec{a}$ and $\vec{b}$ are collinear (unlike because $m=-2<0$)

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198. **Find the direction cosines of the vector $\hat{i}+2\hat{j}+3\hat{k}$.**

Sol: The given vector is $\vec{a}=\hat{i}+2\hat{j}+3\hat{k}\Rightarrow|\vec{a}|=\sqrt{1^2+2^2+3^2}=\sqrt{24}$

$$\hat{a}=\frac{\vec{a}}{|\vec{a}|}=\frac{\hat{i}+2\hat{j}+3\hat{k}}{\sqrt{14}}=\frac{1}{\sqrt{14}}\hat{i}+\frac{2}{\sqrt{14}}\hat{j}+\frac{3}{\sqrt{14}}\hat{k}$$

We know that direction consines of a vectors $\vec{a}$ are coefficients of $\hat{i}$, $\hat{j}$, $\hat{k}$ in $\hat{a}$ $\frac{1}{\sqrt{14}},\frac{2}{\sqrt{14}},\frac{3}{\sqrt{14}}$

$\therefore$ Dc's of the given vector $=\left(\frac{1}{\sqrt{14}},\frac{2}{\sqrt{14}},\frac{3}{\sqrt{14}}\right)$

199. **Show that the vector $\hat{i}+\hat{j}+\hat{k}$ is equally inclined to the axes OX, OY and OZ.**

Sol: Let $\vec{a}=\hat{i}+\hat{j}+\hat{k} \Rightarrow |\vec{a}|=\sqrt{1^2+1^2+1^2}=\sqrt{3}$

DCs of $\vec{a}=\left(\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right)\Rightarrow \cos\alpha=\frac{1}{\sqrt{3}},\cos\beta=\frac{1}{\sqrt{3}},\cos\gamma=\frac{1}{\sqrt{3}}$

Hence, the vector is equally inclined to OX, OY and OZ.

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200. **Show that the direction cosines of a vector equally inclined to the axes OX, OY and OZ are $\pm\left(\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right)$**

Sol: Let the inclination of the equally inclined vector be α.

Then the DCs of the vectors are $\cos\alpha, \cos\alpha, \cos\alpha$

$\therefore \cos^2\alpha+\cos^2\alpha+\cos^2\alpha=1\Rightarrow 3\cos^2\alpha=1\Rightarrow \cos^2\alpha=\frac{1}{3}\Rightarrow \cos\alpha=\pm\frac{1}{\sqrt{3}}$

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201. **Show that the vectors $2\hat{i}-3\hat{j}+4\hat{k}$ and $-4\hat{i}+6\hat{j}-8\hat{k}$ are collinear.**

Sol: Let $\vec{a}=2\hat{i}-3\hat{j}+4\hat{k}$ and $\vec{b}=-4\hat{i}+6\hat{j}-8\hat{k}=-2(2\hat{i}-3\hat{j}+4\hat{k})=-2\vec{a}$

Here, $\vec{b}=-2\vec{a}=m\vec{a}$ where $m=-2<0$

$\therefore$ Vectors $\vec{a}$ and $\vec{b}$ are collinear (unlike because $m=-2<0$)

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202. **Show that the points A(1, 2, 7), B(2, 6, 3) and C(3, 10, –1) are collinear.**

Sol: Given points are A(1, 2, 7), B(2, 6, 3) , C(3, 10, –1)

D.r's of line joining A and B = (1–2, 2–6, 7–3) = (–1,–4, 4)

D.r's of line joining A and C = (1–3, 2–10, 7+1) = (–2,–8, 8)= 2 (–1,–4, 4)

Hence D.r's of AB and AC are proportional. $\therefore$ A,B,C are collinear.

PQ **Show that the points $A(-2\hat{i}+3\hat{j}+5\hat{k})$, $B(\hat{i}+2\hat{j}+3\hat{k})$ and $C(7\hat{i}-\hat{k})$ are collinear.**

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203. **If $\vec{a}=5\hat{i}-\hat{j}-3\hat{k}$ and $\vec{b}=\hat{i}+3\hat{j}-5\hat{k}$, then show that the vectors $\vec{a}+\vec{b}$ and $\vec{a}-\vec{b}$ are perpendicular.**

Sol: $\vec{a}+\vec{b}=(5\hat{i}-\hat{j}-3\hat{k})+(\hat{i}+3\hat{j}-5\hat{k})=6\hat{i}+2\hat{j}-8\hat{k}$

$\vec{a}-\vec{b}=(5\hat{i}-\hat{j}-3\hat{k})-(\hat{i}+3\hat{j}-5\hat{k})=4\hat{i}-4\hat{j}+2\hat{k}$

Now $(\vec{a}+\vec{b}).(\vec{a}-\vec{b})=(6\hat{i}+2\hat{j}-8\hat{k}).(4\hat{i}-4\hat{j}+2\hat{k})=24-8-16=0$

Scalar product of $\vec{a}+\vec{b}$ and $\vec{a}-\vec{b}$ is zero. Hence these are perpendicular vectors.

204. If $\vec{a}=2\hat{i}+2\hat{j}+3\hat{k}$, $\vec{b}=-\hat{i}+2\hat{j}+\hat{k}$ and $\vec{c}=3\hat{i}+\hat{j}$ are such that $\vec{a}+\lambda\vec{b}$ is perpendicular to $\vec{c}$, then find the value of λ.

Sol: $\vec{a}+\lambda\vec{b}=(2\hat{i}+2\hat{j}+3\hat{k})+\lambda(-\hat{i}+2\hat{j}+\hat{k})=(2-\lambda)\hat{i}+(2+2\lambda)\hat{j}+(3+\lambda)\hat{k}$

Also $\vec{c}=3\hat{i}+\hat{j}=3\hat{i}+\hat{j}+0\hat{k}$

Given that $\vec{a}+\lambda\vec{b}$ is perpendicular to $\vec{c}\Rightarrow(\vec{a}+\lambda\vec{b}).\vec{c}=0$

$\Rightarrow(2-\lambda)3+(2+2\lambda)1+(3+\lambda)0=0$

$\Rightarrow 6-3\lambda+2+2\lambda=0\Rightarrow-\lambda+8=0\Rightarrow-\lambda=-8\Rightarrow\lambda=8$

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205. Find the projection of the vector $\vec{a}=2\hat{i}+3\hat{j}+2\hat{k}$ on the vector $\vec{b}=\hat{i}+2\hat{j}+\hat{k}$

Sol: The projection of vector $\vec{a}$ on the vector $\vec{b}$ is given by $\frac{1}{|\vec{b}|}(\vec{a}\cdot\vec{b})$

$$=\frac{(2\hat{i}+3\hat{j}+2\hat{k}).(\hat{i}+2\hat{j}+\hat{k})}{|\hat{i}+2\hat{j}+\hat{k}|}=\frac{(2\times1+3\times2+2\times1)}{\sqrt{(1)^2+(2)^2+(1)^2}}=\frac{10}{\sqrt{6}}=\frac{5}{3}\sqrt{6}$$

PQ Find the projection of the vector $\hat{i}+3\hat{j}+7\hat{k}$ on the vector $7\hat{i}-\hat{j}+8\hat{k}$ [Ans: $\frac{60}{\sqrt{114}}$]

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206. Find $|\vec{a}\times\vec{b}|$, if $\vec{a}=2\hat{i}+\hat{j}+3\hat{k}$ and $\vec{b}=3\hat{i}+5\hat{j}-2\hat{k}$

Sol: We have $\vec{a}\times\vec{b}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\2&1&3\\3&5&-2\end{vmatrix}=\hat{i}(-2-15)-(-4-9)\hat{j}+(10-3)\hat{k}=-17\hat{i}+13\hat{j}+7\hat{k}$

$$\therefore|\vec{a}\times\vec{b}|=\sqrt{(-17)^2+(13)^2+(7)^2}=\sqrt{289+169+49}=\sqrt{507}$$

PQ Find $|\vec{a}\times\vec{b}|$, if $\vec{a}=\hat{i}-7\hat{j}+7\hat{k}$ and $\vec{b}=3\hat{i}-2\hat{j}+2\hat{k}$ [Ans: $19\sqrt{2}$]

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207. Find the area of a parallelogram whose adjacent sides are given by the vectors $\vec{a}=3\hat{i}+\hat{j}+4\hat{k}$ and $\vec{b}=\hat{i}-\hat{j}+\hat{k}$

Sol: The area of a parallelogram with $\vec{a}$ and $\vec{b}$ as its adjacent sides is given by $|\vec{a}\times\vec{b}|$.

Now, $\vec{a}\times\vec{b}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\3&1&4\\1&-1&1\end{vmatrix}=5\hat{i}+\hat{j}-4\hat{k}$

$\therefore|\vec{a}\times\vec{b}|=\sqrt{25+1+16}=\sqrt{42}$.

Thus, the required area is $\sqrt{42}$ sq.units

2 Marks

5 3D-GEOMETRY

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208. **If a line makes angles 90°, 135°, 45° with the x, y and z-axes respectively, find its direction cosines.**

Sol: Let l, m, n be the direction cosines of the given line.

Then $l = \cos 90° = 0$; $m = \cos 135^\circ = -\frac{1}{\sqrt{2}}$; $n = \cos 45^\circ = \frac{1}{\sqrt{2}}$ $\left[\because \cos(90^\circ + 45^\circ) = -\sin 45^\circ\right]$

$\therefore$ the DC's of the line are $0, \frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$

PQ **If a line makes angle 90°, 60° and 30° with the positive direction of x, y and z-axis respectively, find its direction cosines.** [Ans : $0, 1/2, \frac{\sqrt{3}}{2}$]

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209. **If a line makes an angles of 60° and 45° with the direction of positive X and Z aixs respectively. Find the angle made by the line with the Y-aixs.**

Sol: Let l, m , n be the direction cosines of the given line.

The line makes an angle β with y-axis.

Then $l = \cos 60° = \frac{1}{2}$; $m = \cos\beta$; $n = \cos 45° = \frac{1}{\sqrt{2}}$

We know that $l^2 + m^2 + n^2 = 1$

$$\Rightarrow \frac{1}{4} + m^2 + \frac{1}{2} = 1 \Rightarrow m^2 + \frac{1+2}{4} = 1 \Rightarrow m^2 = 1 - \frac{3}{4} = \frac{1}{4} \Rightarrow \cos^2\beta = \frac{1}{4} \Rightarrow \cos\beta = \pm\frac{1}{2}$$

$\therefore \beta = 60°$ (or) $120°$

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210. **Find the direction cosines of a line which makes equal angles with the coordinate axes.**

Sol: Let the line makes an angle α with each of the coordinates axes.

Then $l = \cos\alpha$, $m = \cos\alpha$, $n = \cos\alpha$

We know that $l^2 + m^2 + n^2 = 1 \Rightarrow \cos^2\alpha + \cos^2\alpha + \cos^2\alpha = 1 \Rightarrow 3\cos^2\alpha = 1$

$$\Rightarrow \cos^2\alpha = \frac{1}{3} \Rightarrow \cos\alpha = \pm\frac{1}{\sqrt{3}},$$

$\therefore$ D.C's of the line are $\pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}$

211. Find the direction cosines of the joining the points (–2, 6, 7) and (1, 2, –5).

Sol: Let $A(x_1,y_1,z_1)=(-2,6,7)$, $B(x_2,y_2,z_2)=(1,2,-5)$

D.r's of $AB=(a,b,c)=(x_2-x_1,y_2-y_1,z_2-z_1)=(1+2,2-6,-5-7)=(3,-4,-12)$

$\therefore \sqrt{a^2+b^2+c^2}=\sqrt{9+16+144}=\sqrt{169}=13$

DC's of $AB=\left(\frac{a}{\sqrt{a^2+b^2+c^2}},\frac{b}{\sqrt{a^2+b^2+c^2}},\frac{c}{\sqrt{a^2+b^2+c^2}}\right)=\left(\frac{3}{13},\frac{-4}{13},\frac{-12}{13}\right)$

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212. Show that the points (2, 3, 4), (– 1, – 2, 1), (5, 8, 7) are collinear.

Sol: Given points are A(2, 3, 4), B(– 1, – 2, 1), C(5, 8, 7).

D.r's of line joining A and B = (–1–2, –2–3, 1–4) = (–3,–5, –3) = (3,5,3)

D.r's of line joining A and C = (5–2, 8–3, 7–4) = (3,5, 3)

Hence D.r's of AB and AC are proportional. $\therefore$ A,B,C are collinear.

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213. If the points (2, –1, 3), (4, *a*, 1) and (3, 1, *b*) are collinear, then find the ratio between *a* and *b*?

Sol: Let $A(x_1,y_1,z_1)=(2,-1,3)$, $B(x_2,y_2,z_2)=(4,a,1)$, $C(x_3,y_3,z_3)=(3,1,b)$ **[BMP]**

If A,B,C are collinear then $\frac{x_1-x_2}{x_2-x_3}=\frac{y_1-y_2}{y_2-y_3}=\frac{z_1-z_2}{z_2-z_3}$

$\Rightarrow \frac{2-4}{4-3}=\frac{-1-a}{a-1}=\frac{3-1}{1-b}\Rightarrow \frac{-2}{1}=\frac{-1-a}{a-1}=\frac{2}{1-b}$

$\Rightarrow \frac{-2}{1}=\frac{-1-a}{a-1}\Rightarrow 2a-2=1+a\Rightarrow 2a-a=1+2\Rightarrow a=3$

$\Rightarrow \frac{-2}{1}=\frac{2}{1-b}\Rightarrow \frac{-1}{1}=\frac{1}{1-b}\Rightarrow -1+b=1\Rightarrow b=2$ $\therefore$ Required ratio $\frac{a}{b}=\frac{3}{2}\Rightarrow 3:2$

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214. Find the equation of the line which passes through the point (1, 2, 3) and is parallel to the vector $3\hat{i}+2\hat{j}-2\hat{k}$.

Sol: Cartesian form with $(x_1,y_1,z_1)=(1,2,3)$ and the direction ratios (a, b, c)= (3,2,–2) is

$\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-z_1}{c}\Rightarrow \frac{x-1}{3}=\frac{y-2}{2}=\frac{z-3}{-2}$

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215. Find the angle between the following pairs of lines:
$\vec{r}=2\hat{i}-5\hat{j}+\hat{k}+\lambda(3\hat{i}+2\hat{j}+6\hat{k})$ and $\vec{r}=7\hat{i}-6\hat{k}+\mu(\hat{i}+2\hat{j}+2\hat{k})$

Sol: Required angle is the angle between $\vec{b_1}=3\hat{i}+2\hat{j}+6\hat{k}$ and $\vec{b_2}=\hat{i}+2\hat{j}+2\hat{k}$

$|\vec{b_1}|=\sqrt{3^2+2^2+6^2}=\sqrt{49}=7$; $|\vec{b_2}|=\sqrt{1^2+2^2+2^2}=\sqrt{9}=3$

$\vec{b_1}.\vec{b_2}=(3i+2j+6k).(i+2j+2k)=3(1)+2(2)+6(2)=3+4+12=19$

$\therefore \cos\theta=\left|\frac{\vec{b_1}.\vec{b_2}}{|\vec{b_1}||\vec{b_2}|}\right|=\left|\frac{19}{7\times 3}\right|=\frac{19}{21}\Rightarrow\theta=\cos^{-1}\left(\frac{19}{21}\right)$

PQ **Find the angle between the pair of lines given by** $\vec{r}=3\hat{i}+2\hat{j}-4\hat{k}+\lambda(\hat{i}+2\hat{j}+2\hat{k})$ **and** $\vec{r}=5\hat{i}-2\hat{j}+\mu(3\hat{i}+2\hat{j}+6\hat{k})$ $\left[\text{Ans}:\theta=\cos^{-1}\left(\frac{19}{21}\right)\right]$

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216. **Find the angle between the following pair of lines:**

$\frac{x-2}{2}=\frac{y-1}{5}=\frac{z+3}{-3}$ and $\frac{x+2}{-1}=\frac{y-4}{8}=\frac{z-5}{4}$

Sol: Reducing the pair of lines into vector form $\vec{b_1}=2\hat{i}+5\hat{j}-3\hat{k}$ and $\vec{b_2}=-\hat{i}+8\hat{j}+4\hat{k}$

$|\vec{b_1}|=\sqrt{(2)^2+(5)^2+(-3)^2}=\sqrt{38}$; $|\vec{b_2}|=\sqrt{(-1)^2+(8)^2+(4)^2}=\sqrt{81}=9$

$\vec{b_1}.\vec{b_2}=(2\hat{i}+5\hat{j}-3\hat{k}).(-\hat{i}+8\hat{j}+4\hat{k})=2(-1)+5\times8+(-3)4=-2+40-12=26$

$\therefore\cos\theta=\left|\frac{\vec{b_1}.\vec{b_2}}{|\vec{b_1}||\vec{b_2}|}\right|=\left|\frac{26}{\sqrt{38}\times9}\right|=\frac{26}{9\sqrt{38}}\Rightarrow\theta=\cos^{-1}\left(\frac{26}{9\sqrt{38}}\right)$

PQ **Find the angle between the pair of lines** $\frac{x+3}{3}=\frac{y-1}{5}=\frac{z+3}{4}$ and $\frac{x+1}{1}=\frac{y-4}{1}=\frac{z-5}{2}$ $\left[\text{Ans}:\theta=\cos^{-1}\left(\frac{8\sqrt{3}}{15}\right)\right]$

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217. **Show that the lines** $\frac{x-5}{7}=\frac{y+2}{-5}=\frac{z}{1}$ and $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$ **are perpendicular to each other.**

Sol: From the given lines we have $\vec{b_1}=7\hat{i}-5\hat{j}+\hat{k}$ and $\vec{b_2}=\hat{i}+2\hat{j}+3\hat{k}$

$\vec{b_1}.\vec{b_2}=(7\hat{i}-5\hat{j}+\hat{k}).(\hat{i}+2\hat{j}+3\hat{k})=7(1)-5\times2+1(3)=7-10+3=0$

$\therefore$ the given lines are perpendicular to each other.

PQ **Show that the pair of lines** $\frac{4-x}{3}=\frac{2y+1}{2}=\frac{z-1}{5}$ and $\frac{2x+1}{2}=\frac{3-y}{2}=\frac{3z-1}{3}$ **are perpendicular to each other.**

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218. **If the lines** $\frac{x-1}{-3}=\frac{y-2}{2k}=\frac{z-3}{2}$ and $\frac{x-1}{3k}=\frac{y-1}{1}=\frac{z-6}{-5}$ **are perpendicular, find the value of k.**

Sol: From the given lines we have $\vec{b_1}=-3\hat{i}+2k\hat{j}+2\hat{k}$ and $\vec{b_2}=3k\hat{i}+\hat{j}-5\hat{k}$

When the lines are perpendicular then $\vec{b_1}.\vec{b_2}=0$

$\vec{b_1}.\vec{b_2}=(-3\hat{i}+2k\hat{j}+2\hat{k}).(3k\hat{i}+\hat{j}-5\hat{k})=0\Rightarrow-3(3k)+2k(1)+2(-5)=0$

$\Rightarrow-9k+2k-10=0\Rightarrow7k=-10\Rightarrow k=\frac{-10}{7}$

PQ **Find the values of p so that the lines**

$\frac{1-x}{3}=\frac{7y-14}{2p}=\frac{z-3}{2}$ and $\frac{7-7x}{3p}=\frac{y-5}{1}=\frac{6-z}{5}$ **are at right angles.** $\left[\text{Ans}:\ p=\frac{70}{11}\right]$

2 Marks

6 PROBABILITY

FIRST SIX ARE BEST SIX

219. If $P(A)=\frac{7}{13}$, $P(B)=\frac{9}{13}$ and $P(A\cap B)=\frac{4}{13}$, evaluate $P(A|B)$.

Sol: We have $P(A|B)=\frac{P(A\cap B)}{P(B)}=\frac{\frac{4}{13}}{\frac{9}{13}}=\frac{4}{\cancel{13}}\times\frac{\cancel{13}}{9}=\frac{4}{9}$

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220. Compute $P(A|B)$, if $P(B)=0.5$ and $P(A\cap B)=0.32$

Sol: Given that $P(B)=0.5$ and $P(A\cap B)=0.32$ $\quad\therefore P(A/B)=\frac{P(A\cap B)}{P(B)}=\frac{0.32}{0.5}=\frac{32}{50}=\frac{16}{25}$

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221. If $P(A)=\frac{3}{5}$ and $P(B)=\frac{1}{5}$, find $P(A\cap B)$ if A and B are independent events.

Sol: Given that $P(A)=\frac{3}{5}$, $P(B)=\frac{1}{5}$. Also A,B are independent events.

$\therefore P(A\cap B)=P(A)P(B)=\frac{3}{5}\times\frac{1}{5}=\frac{3}{25}$

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222. A and B are two events such that $P(A)\neq 0$. Find $P(B|A)$, if **[BMP]**
(i) A is a subset of B (ii) $A\cap B=\phi$

Sol: (i) A is a subset of $B\Rightarrow A\cap B=A$. Then $P(A\cap B)=P(B\cap A)=P(A)$

$\therefore P(B|A)=\frac{P(B\cap A)}{P(A)}=\frac{P(A)}{P(A)}=1$

(ii) $A\cap B=\phi\Rightarrow P(A\cap B)=0\quad\therefore P(B|A)=\frac{P(A\cap B)}{P(A)}=\frac{0}{P(A)}=0,\quad\because P(A)\neq 0$

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223. Given that E and F are events such that $P(E)=0.6$, $P(F)=0.3$ and $P(E\cap F)=0.2$, find $P(E|F)$ and $P(F|E)$

Sol: Given that $P(E)=0.6$, $P(F)=0.3$, $P(E\cap F)=0.2$

$\therefore P(E/F)=\frac{P(E\cap F)}{P(F)}=\frac{0.2}{0.3}=\frac{2}{3}$ and $P(F/E)=\frac{P(F\cap E)}{P(E)}=\frac{0.2}{0.6}=\frac{1}{3}$.

224. **Evaluate $P(A \cup B)$, if $2P(A) = P(B) = \frac{5}{13}$ and $P(A|B) = \frac{2}{5}$**

Sol: Given that $2P(A) = P(B) = \frac{5}{13} \Rightarrow P(A) = \frac{5}{26}, P(B) = \frac{5}{13}$

Now $P(A|B) = \frac{2}{5} \Rightarrow \frac{P(A \cap B)}{P(B)} = \frac{2}{5} \Rightarrow P(A \cap B) = \frac{2}{5} P(B) = \frac{2}{\cancel{5}} \times \frac{\cancel{5}}{13} = \frac{2}{13}$

$\therefore P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{5}{26} + \frac{5}{13} - \frac{2}{13} = \frac{5+10-4}{26} = \frac{11}{26}$.

---

225. **Determine P(E|F) when a coin is tossed three times, where E : at least two heads , F : at most two heads**

Sol: **E : at least two heads** $\Rightarrow$ E={HHH, HHT, HTH, THH} $\Rightarrow$ n(E) = 4
F : at most two heads $\Rightarrow$ F={HHT, HTH, THH, HTT, THT, TTH, TTT} $\Rightarrow$ n(F) =7
Hence $E \cap F$ = {HHT, HTH, THH} $\Rightarrow$ $n(E \cap F) = 3$

Now $P(E) = \frac{4}{8} = \frac{1}{2}, P(F) = \frac{7}{8}, \quad P(E \cap F) = \frac{n(E \cap F)}{n(S)} = \frac{3}{8}$

$\therefore P(E/F) = \frac{P(E \cap F)}{P(F)} = \frac{3/8}{7/8} = \frac{3}{\cancel{8}} \times \frac{\cancel{8}}{7} = \frac{3}{7}$.

PQ **Determine P(E|F) when a coin is tossed three times, where E : at most two tails , F : at least one tail.** [Ans: 6/7]

---

226. **An urn contains 10 black and 5 white balls. Two balls are drawn from the urn one after the other without replacement. What is the probability that both drawn balls are black?**

Sol: Let A and B denote the events that first and second ball drawn are black.

The probability of getting a black ball in the first draw is $P(A) = \frac{10}{15}$

Probability of getting a black ball in the second drawn given first drawn is black $P(B|A) = \frac{9}{14}$

$P(A \cap B) = P(A)\, P(B|A) = \frac{10}{15} \times \frac{9}{14} = \frac{3}{7}$ ($\because$ first drawn ball is not replaced)

227. **Two cards are drawn at random in succession and without replacement from a pack of 52 playing cards. Find the probability that both the cards are black.**

Sol: There are 26 black cards in a deck of 52 cards.

Let A,B be the events of drawing black cards in the successive draws.

Probability of getting a black card in the first draw is $P(A)=\frac{26}{52}$

Probability of getting a black card in the second drawn given first drawn is black $P(B|A)=\frac{25}{51}$

$\therefore P(A\cap B)=P(A)P(B|A)=\frac{26}{52}\times\frac{25}{51}=\frac{25}{102}$ ($\because$ first drawn ball is not replaced)

---

228. **A family has two children. What is the probability that both the children are boys given that at least one of them is a boy ?**

Sol: Let b stand for boy and g for girl.

The sample space is S = {(b, b), (g, b), (b, g), (g, g)}

Let E and F denote the following events :

E : 'both the children are boys' then E = {(b,b)}

F : 'at least one of the child is a boy' then F = {(b,b), (g,b), (b,g)}

Now $E\cap F=\{(b,b)\}$. Thus $P(F)=\frac{3}{4}$ and $P(E\cap F)=\frac{1}{4}$

$\therefore P(E|F)=\frac{P(E\cap F)}{P(F)}=\frac{\frac{1}{4}}{\frac{3}{4}}=\frac{1}{\cancel{4}}\times\frac{\cancel{4}}{3}=\frac{1}{3}$

---

229. **If a leap year is selected at random, what is the chance that it will contain 53 tuesdays?**

Sol: In a leap year, there are 366 days i.e., 52 weeks and 2 days.

In 52 weeks, there are 52 Tuesdays.

$\therefore$ the probability that the leap year will contain 53 Tuesdays is equal to the probability that the remaining 2 days will be Tuesdays.

The remaining 2 days can be any of the following:

(Mon & Tue), (Tue & Wed), (Wed & Thu), (Thu & Fri), (Fri & Sat), (Sat & Sun) , (Sun & Mon)

Total number of cases =7; Favourable cases=2

Probability that a leap year will have 53 Tuesdays =2/7

2 Marks

7 CONTINUITY & DIFFERENTIABILITY

MANY STUDENTS LIKE DOING DIFFERENTIATION

230. **Differentiate the function $\sin(x^2+5)$ with respect to x .**

Sol: Let $y=\sin(x^2+5)$

$$\therefore \frac{dy}{dx}=\frac{d}{dx}\sin(x^2+5)=\cos(x^2+5)\frac{d}{dx}(x^2+5) \quad \left[\because \frac{d}{dx}\sin f(x)=\cos f(x)\frac{d}{dx}f(x)\right]$$

$$=\cos(x^2+5)(2x+0)=2x\cos(x^2+5). \quad \left[\because \frac{d}{dx}x^n=nx^{n-1} \text{ and } \frac{d}{dx}(c)=0\right]$$

---

231. **Differentiate the function $\cos(\sqrt{x})$ with respect to x .**

Sol: Let $y=\cos(\sqrt{x})$

$$\therefore \frac{dy}{dx}=\frac{d}{dx}\cos(\sqrt{x})=-\sin\sqrt{x}\frac{d}{dx}\sqrt{x}=-\sin\sqrt{x}\frac{1}{2\sqrt{x}} \quad \left[\because \frac{d}{dx}\sqrt{x}=\frac{1}{2\sqrt{x}}\right]$$

---

232. **Differentiate the function $\cos(\sin x)$ with respect to x .** **[BMP]**

Sol: Let $y=\cos(\sin x)$

$$\therefore \frac{dy}{dx}=\frac{d}{dx}\cos(\sin x)=-\sin(\sin x)\frac{d}{dx}(\sin x) \quad \left[\because \frac{d}{dx}\cos f(x)=-\sin f(x)\frac{d}{dx}f(x)\right]$$

$$=-\sin(\sin x)\cos x=-\cos x\sin(\sin x)$$

PQ **Differentiate sin (log x), x > 0 w.r.t. x.** $\left[\text{Ans}:\frac{\cos(\log x)}{x}\right]$

---

233. **Differentiate log (log x), x>1 w.r.t. x.**

Sol: Let $y=\log(\log x)$

$$\therefore \frac{dy}{dx}=\frac{1}{\log x}\frac{d}{dx}(\log x)=\frac{1}{\log x}\frac{1}{x}=\frac{1}{x\log x}. \quad \left[\because \frac{d}{dx}\log f(x)=\frac{1}{f(x)}\frac{d}{dx}f(x)\right]$$

---

234. **Differentiate e^{x^3} w.r.t. x.**

Sol: Let $y=e^{x^3}=e^{(x^3)}$

$$\therefore \frac{dy}{dx}=e^{(x^3)}\frac{d}{dx}x^3=e^{(x^3)}3x^2=3x^2e^{(x^3)} \quad \left[\because \frac{d}{dx}e^{f(x)}=e^{f(x)}\frac{d}{dx}f(x)\right]$$

PQ **Differentiate $e^{\sin^{-1}x}$ w.r.t. x .** $\left[\text{Ans}: e^{\sin^{-1}x}\cdot\frac{1}{\sqrt{1-x^2}}\right]$

235. Find $\frac{dy}{dx}$ in $y=\sin^{-1}\left(\frac{2x}{1+x^2}\right)$

Sol: Put $x=\tan\theta$ then we have $\theta=\tan^{-1}x$

$$\therefore\ y=\sin^{-1}\left(\frac{2\tan\theta}{1+\tan^2\theta}\right)=\sin^{-1}(\sin 2\theta)=2\theta=2\tan^{-1}x$$

$$\therefore \frac{dy}{dx}=2\left(\frac{1}{1+x^2}\right)=\frac{2}{1+x^2}$$

PQ Find $\frac{dy}{dx}$ in $y=\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right), 0<x<1$ $\left[\text{Ans}: \frac{dy}{dx}=\frac{2}{1+x^2}\right]$

---

236. Find $\frac{dy}{dx}$ in $y=\tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right), -\frac{1}{\sqrt{3}}<x<\frac{1}{\sqrt{3}}$

Sol: Put $x=\tan\theta$ then we have $\theta=\tan^{-1}x$

$$\therefore y=\tan^{-1}\left(\frac{3\tan\theta-\tan^3\theta}{1-3\tan^2\theta}\right)=\tan^{-1}(\tan 3\theta)=3\theta=3\tan^{-1}x\ \therefore \frac{dy}{dx}=3\left(\frac{1}{1+x^2}\right)=\frac{3}{1+x^2}$$

---

237. Find $\frac{dy}{dx}$ of $x^2+xy+y^2=100$

Sol: Given that $x^2+xy+y^2=100 \Rightarrow \frac{d}{dx}x^2+\frac{d}{dx}xy+\frac{d}{dx}y^2=\frac{d}{dx}(100)$

$$\therefore\ 2x+\left(x\frac{d}{dx}y+y\frac{d}{dx}x\right)+2y\frac{dy}{dx}=0 \Rightarrow 2x+x\frac{dy}{dx}+y+2y\frac{dy}{dx}=0$$

$$\Rightarrow (x+2y)\frac{dy}{dx}=-2x-y \Rightarrow \frac{dy}{dx}=-\frac{(2x+y)}{x+2y}.$$

---

238. Find $\frac{dy}{dx}$, if $x=at^2, y=2at$.

Sol: Given that $x=at^2$, $y=2at$. Then $\frac{dx}{dt}=2at$ and $\frac{dy}{dt}=2a$

$$\therefore \frac{dy}{dx}=\frac{\frac{dy}{dt}}{\frac{dx}{dt}}=\frac{2a}{2at}=\frac{1}{t}$$

239. Find $\frac{dy}{dx}$ of $x = a\cos\theta, y = b\cos\theta$

Sol: Given that $x = a\cos\theta$ and $y = b\cos\theta$. Differentiating both eqns. w.r.t. θ have

$\frac{dx}{d\theta} = \frac{d}{d\theta}(a\cos\theta) = a\frac{d}{d\theta}\cos\theta = -a\sin\theta$ and $\frac{dy}{d\theta} = \frac{d}{d\theta}(b\cos\theta) = b\frac{d}{d\theta}\cos\theta = -b\sin\theta$

$\therefore \frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{-b\sin\theta}{-a\sin\theta} = \frac{b}{a}$.

PQ **Find $\frac{dy}{dx}$ of $x = a\sec\theta, y = b\tan\theta$** $\left[\text{Ans}: \frac{b}{a}\operatorname{cosec}\theta\right]$

---

240. **Find $\frac{dy}{dx}$, if $x = a(\theta + \sin\theta), y = a(1 - \cos\theta)$.**

Sol: Given that $x = a(\theta + \sin\theta)$ and $y = a(1 - \cos\theta)$. Then $\frac{dx}{d\theta} = a(1+\cos\theta), \frac{dy}{d\theta} = a(\sin\theta)$

$$\therefore \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{a\sin\theta}{a(1+\cos\theta)} = \frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\cos^2\frac{\theta}{2}} = \tan\frac{\theta}{2}$$

PQ **Find $\frac{dy}{dx}$ of $x = a(\theta - \sin\theta), y = a(1 + \cos\theta)$** $\left[\text{Ans}: -\cot\frac{\theta}{2}\right]$

---

241. **Find the second order derivative of $x^2 + 3x + 2$.**

Sol: Let $y = x^2 + 3x + 2 \Rightarrow \frac{dy}{dx} = 2x + 3(1) + 0 = 2x + 3$. Again differentiating w.r.t x, we have

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = 2(1) + 0 = 2$$

PQ **Find the second order derivative of x^{20}** $\left[\text{Ans}: 380x^{18}\right]$

---

242. **Find the second order derivative of log x**

Sol: Let $y = \log x \Rightarrow \frac{dy}{dx} = \frac{1}{x}$ Again differentiating w.r.t x, we have

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{1}{x}\right) = \frac{d}{dx}(x^{-1}) = (-1)x^{-2} = \frac{-1}{x^2}.$$

---

243. **If $y = A\sin x + B\cos x$, then prove that $\frac{d^2y}{dx^2} + y = 0$**

Sol: Given that $y = A\sin x + B\cos x \Rightarrow \frac{dy}{dx} = A\cos x - B\sin x$ and

$$\frac{d^2y}{dx^2} = \frac{d}{dx}(A\cos x - B\sin x) = -A\sin x - B\cos x = -y \Rightarrow \frac{d^2y}{dx^2} + y = 0$$

2 Marks

8 APPLICATIONS OF DERIVATIVES

ALL EASY APPLICATIONS IN JUST 2 PAGES

244. **Show that the function given by $f(x) = 3x + 17$ is increasing on R.**

Sol: Given $f(x) = 3x + 17 \Rightarrow f'(x) = 3 > 0 \ \forall x \in R$. $\therefore f(x)$ is increasing on R.

PQ **Show that the function given by $f(x) = 7x - 3$ is increasing on R.**

---

245. **Show that the function given by $f(x) = e^{2x}$ is increasing on R.**

Sol: Given $f(x) = e^{2x} \Rightarrow f'(x) = e^{2x}(2) > 0 \ \forall x \in R$. $\therefore f(x)$ is increasing on R.

PQ **Prove that the logarithmic function is increasing on $(0, \infty)$.**

---

246. **Show that the function f given by $f(x) = x^3 - 3x^2 + 4x$, $x \in R$ is increasing on R.** **[BMP]**

Sol: Given that $f(x) = x^3 - 3x^2 + 4x$

$\Rightarrow f'(x) = 3x^2 - 6x + 4 = 3(x^2 - 2x + 1) + 1 = 3(x - 1)^2 + 1 > 0 \ \forall x \in R$

$\therefore f(x)$ is increasing on **R**.

PQ **Prove that the function given by $f(x) = x^3 - 3x^2 + 3x - 100$ is increasing in R.**

---

247. **Prove that the function given by $f(x) = \cos x$ is decreasing in $(0, \pi)$**

Sol: Given that $f(x) = \cos x \Rightarrow f'(x) = -\sin x < 0 \ \forall x \in (0, \pi)$ $[\because \sin x > 0$ for $x \in (0, \pi)]$

$\therefore f(x)$ is is decreasing in $(0, \pi)$.

PQ **Show that the function given by $f(x) = \sin x$ isincreasing in $\left(0, \frac{\pi}{2}\right)$**

---

248. **Find the intervals in which the function f given by $f(x) = x^2 - 4x + 6$ is**

(a) increasing (b) decreasing

Sol: Given that $f(x) = x^2 - 4x + 6 \Rightarrow f'(x) = 2x - 4 = 2(x - 2)$

(i) $f(x)$ is increasing when $f'(x) > 0 \Rightarrow 2(x-2) > 0 \Rightarrow x - 2 > 0 \Rightarrow x > 2 \Rightarrow x \in (2, \infty)$

(ii) $f(x)$ is decreasing when $f'(x) < 0 \Rightarrow 2(x-2) < 0 \Rightarrow x < 2 \Rightarrow x \in (-\infty, 2)$

---

249. **Find the intervals in which the function $x^2 + 2x - 5$ is strictly increasing or decreasing**

Sol: Let $f(x) = x^2 + 2x - 5 \Rightarrow f'(x) = 2x + 2 = 2(x + 1)$

(i) $f(x)$ is increasing when $f'(x) > 0 \Rightarrow x + 1 > 0 \Rightarrow x > -1 \Rightarrow x \in (-1, \infty)$

(ii) $f(x)$ is decreasing when $f'(x) < 0 \Rightarrow x + 1 < 0 \Rightarrow x < -1 \Rightarrow x \in (-\infty, -1)$.

250. **For what values of a the function f given by $f(x) = x^2 + ax + 1$ is increasing on [1, 2]?**

Sol: Given that $f(x) = x^2 + ax + 1 \Rightarrow f'(x) = 2x + a$.

Now, f has to be increasing on [1,2]. Also 2x+a is a linear function.

In [1,2] its minimum occurs at x=1 $\therefore$ minimum value $=2(1)+a=2+a$

Since, the function is increasing, $2 + a \geq 0 \Rightarrow a \geq -2$

PQ **Find the intervals in which the function f given by $f(x) = 2x^3 - 3x^2 - 36x + 7$ is (a) increasing (b) decreasing [A:** Increasing in $(-\infty,-2)\cup(3,\infty)$; decreasing in $(-2,3)$]

---

251. **Find the maximum and minimum values of $f(x) = 9x^2 + 12x + 2$**

Sol: Given that $f(x) = 9x^2 + 12x + 2 \Rightarrow f'(x) = 18x+12=6(3x+2)$

$\Rightarrow f'(x) = 18>0$ $\therefore$ f(x) has only minimum value but no maximum value.

At extreme values $f'(x)=0 \Rightarrow 3x+2=0 \Rightarrow 3x+2=0 \Rightarrow x=-\frac{2}{3}$

Minimum value of f is $f\left(-\frac{2}{3}\right)=9\left(\frac{-2}{3}\right)^2+12\left(-\frac{2}{3}\right)+2=9\left(\frac{4}{9}\right)-8+2=-2$

---

252. **Find the rate of change of the area of a circle per second with respect to its radius r when r = 5 cm.**

Sol: For the circle, we take radius = r and area =A

Area of circle $A = \pi r^2 \Rightarrow A = \pi r^2 \Rightarrow \frac{dA}{dr} = \pi(2r) = 2\pi r$ When $r = 5, \frac{dA}{dr} = 2\pi(5) = 10\pi.$ cm^2

Thus, the area of the circle is changing at the rate of 10π cm^2/s.

---

253. **Find the rate of change of the area of a circle with respect to its radius r when r = 3 cm** **[BMP]**

Sol: For the circle, we take radius = r and area =A Given r= 3 cm

Area of circle $A = \pi r^2 \Rightarrow A = \pi r^2 \Rightarrow \frac{dA}{dr} = \pi(2r) = 2\pi r$

When $r = 3, \frac{dA}{dr} = 2\pi(3) = 6\pi.$

Thus, the area of the circle is changing at the rate of 6π cm^2/s

---

254. **The radius of a circle is increasing uniformly at the rate of 3 cm/s. Find the rate at which the area of the circle is increasing when the radius is 10 cm.**

Sol: For the circle, we take radius = r and area =A. Area of circle $A=\pi r^2$.

Given $\frac{dr}{dt} = 3cm/s$ and r= 10 cm $\Rightarrow \frac{dA}{dt} = \frac{d}{dt}(\pi r^2) = 2\pi r\frac{dr}{dt} = 2\pi(10)(3) = 60\pi$ cm^2/s

PQ **The radius of an air bubble is increasing at the rate of 1/2 cm/s. At what rate is the volume of the bubble increasing when the radius is 1 cm? [A: 2π cm^3/s]**

2 Marks

9 INTEGRATION

ALL EASY INTEGRATION PROBLEMS ARE HERE FOR YOU

PROBLEMS ON INDEFINITE INTEGRALS

255. Find $\int(ax^2+bx+c)dx$

Sol: $\int(ax^2+bx+c)dx=\int ax^2\,dx+\int bx\,dx+\int c\,dx$

$=a\int x^2dx+b\int x^1dx+c\int 1dx \quad =a\frac{x^3}{3}+b\frac{x^2}{2}+cx+c_1$ where c_1 is the constant of integration.

---

256. Find integral of $\frac{2x}{1+x^2}$

Sol: Put $1+x^2=t\Rightarrow 2xdx=dt$

$\therefore\int\frac{2x}{1+x^2}dx=\int\frac{dt}{t}=\int\frac{1}{t}dt=\log|t|+c=\log|1+x^2|+c=\log(1+x^2)+c.\quad[\because t=1+x^2]$

---

257. Find the integral of $\frac{3x^2}{x^6+1}$

Sol: Put $x^3=t\Rightarrow 3x^2dx=dt$

$\therefore\int\frac{3x^2}{x^6+1}dx=\int\frac{dt}{t^2+1}=\tan^{-1}t+C=\tan^{-1}(x^3)+C \qquad \left[\because\int\frac{1}{x^2+a^2}dx=\frac{1}{a}\tan^{-1}\frac{x}{a}+C\right]$

PQ Find integral of $\frac{x^3}{\sqrt{1-x^8}}$ $\left[\text{Hint}: x^4=t,\ \text{Ans}:\frac{1}{4}\sin^{-1}(x^4)+C\right]$

---

258. Find $\int\sec x(\sec x+\tan x)dx$

Sol: $\int\sec x(\sec x+\tan x)dx=\int(\sec^2 x+\sec x\tan x)dx$

$=\int\sec^2 xdx+\int\sec x\tan xdx \quad=\tan x+\sec x+c.$

PQ Find the $\int\text{cosec}x(\text{cosec}x+\cot x)dx$ **[BMP]** $[\text{Ans}:-\cot x-\text{cosec}x+C]$

259. **Find integral of** $\frac{\sin x}{(1+\cos x)^2}$

Sol: Put $1+\cos x=t \Rightarrow -\sin x dx=dt$

$$\therefore \int \frac{\sin x}{(1+\cos x)^2}dx = -\int \frac{dt}{t^2} = \frac{1}{t}+C = \frac{1}{(1+\cos x)}+C \qquad \left(\because \frac{d}{dx}\left(\frac{1}{x^2}\right)=\frac{-1}{x}\right)$$

PQ **Find integral of** $\frac{\sin x}{1+\cos x}$ **[Hint:** $1+\cos x=t$ **Ans:** $-\log|1+\cos x|+C$ **]**

PQ **Find integral of** $\frac{\cos x}{\sqrt{1+\sin x}}$ [Hint : $1+\sin x = t$, Ans : $2\sqrt{1+\sin x}+C$]

---

260. **Find integral of** $\frac{e^{\tan^{-1}x}}{1+x^2}$

Sol: Put $\tan^{-1}x = t \Rightarrow \frac{1}{1+x^2}dx = dt$ $\qquad \therefore \int \frac{e^{\tan^{-1}x}}{1+x^2}dx = \int e^t dt = e^t + C = e^{\tan^{-1}x}+C$

PQ **Find integral of** $\frac{\sin^{-1}x}{\sqrt{1-x^2}}$ $\left[\text{Hint}: \sin^{-1}x = t,\ \text{Ans}: \frac{(\sin^{-1}x)^2}{2}+C\right]$

---

261. **Find integral of** $\frac{\cos\sqrt{x}}{\sqrt{x}}$

Sol: Put $\sqrt{x}=t \Rightarrow \frac{1}{2\sqrt{x}}dx = dt \Rightarrow \int \frac{\cos\sqrt{x}}{\sqrt{x}}dx = 2\int \cos t dt = 2\sin t + C = 2\sin\sqrt{x}+C$

---

262. **Find integral of** $\frac{(\log x)^2}{x}$ **[BMP]**

Sol: Put $\log x=t \Rightarrow \frac{1}{x}dx = dt \Rightarrow \frac{dx}{x} = dt$

$$\therefore \int \frac{(\log x)^2}{x}dx = \int (\log x)^2\left(\frac{dx}{x}\right) = \int t^2 dt = \frac{t^3}{3}+c = \frac{1}{3}(\log x)^3 + c \qquad [\because t = \log x]$$

PQ **Find integral of** $\frac{(1+\log x)^2}{x}$

---

263. **Find the integral of** $e^x(\sin x+\cos x)$

Sol: We know that $\int e^x[f(x)+f'(x)]dx = e^x f(x)+C$. Here $f(x)=\sin x$ and $f'(x)=\cos x$

$$\therefore I = \int e^x(\sin x+\cos x)dx = e^x \sin x + C$$

264. Find the integral of $e^x\left(\frac{1}{x}-\frac{1}{x^2}\right)$

Sol: We know that $\int e^x[f(x)+f'(x)]dx = e^x f(x)+C$; Here, $f(x)=\frac{1}{x}$ and $f'(x)=\frac{-1}{x^2}$

$$\therefore I=\int e^x\left(\frac{1}{x}-\frac{1}{x^2}\right)=e^x\left(\frac{1}{x}\right)+C=\frac{e^x}{x}+C$$

---

265. Find $\int \log x dx$

Sol: We take log x as the first function (u) and 1 as the second function(v) are intergal by parts.

$$\int(\log x.1)dx = \log x\int 1dx-\int[\frac{d}{dx}(\log x)\int 1dx]dx = (\log x)\cdot x-\int\frac{1}{x}xdx = x\log x-x+C$$

---

266. Find the integral of $x \sin x$

Sol: Let $I=\int x\sin x dx$ Taking u=x, v = sinx and integrating by parts, we have

$$I=x\int\sin x dx-\int\left[\left(\frac{d}{dx}(x)\right)\int\sin x dx\right]dx = x(-\cos x)-\int 1\cdot(-\cos x)dx$$

$$= -x\cos x+\sin x+C$$

---

267. Find the integral of $\frac{1}{x^2-9}$

Sol: $$\int\frac{dx}{x^2-9}=\int\frac{dx}{x^2-3^2} = \frac{1}{2\times 3}\log\left|\frac{x-3}{x+3}\right|+C=\frac{1}{6}\log\left|\frac{x-3}{x+3}\right|+C$$

PQ Find $\int\frac{dx}{x^2-16}$ $\left[\text{Ans}: \frac{1}{8}\log\left|\frac{x-4}{x+4}\right|+C\right]$

---

268. Find the integral of $\frac{1}{\sqrt{9-25x^2}}$

Sol: Put $5x=t \Rightarrow 5dx=dt$ $\left[\because\int\frac{1}{\sqrt{a^2-x^2}}dx=\text{Sin}^{-1}\left(\frac{x}{a}\right)+C\right]$

$$\therefore\int\frac{1}{\sqrt{9-25x^2}}dx=\frac{1}{5}\int\frac{1}{\sqrt{9-t^2}}dt=\frac{1}{5}\int\frac{1}{\sqrt{3^2-t^2}}dt=\frac{1}{5}\sin^{-1}\left(\frac{t}{3}\right)+C=\frac{1}{5}\sin^{-1}\left(\frac{5x}{3}\right)+C$$

PROBLEMS ON DEFINITE INTEGRALS

269. Evaluate $\int_2^3 x^2 dx$

Sol: $I=\int_2^3 x^2 dx=\left[\frac{x^3}{3}\right]_2^3=\frac{3^3}{3}-\frac{2^3}{3}=9-\frac{8}{3}=\frac{27-8}{3}=\frac{19}{3}$

PQ Evaluate $\int_1^2 x^5 dx$ [Ans:21/2]

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270. Evaluate $\int_4^5 e^x dx$

Sol: Let $I=\int_4^5 e^x dx=\left[e^x\right]_4^5=e^5-e^4=e^4(e-1)$

PQ Find $\int_0^1 e^x dx$ [Ans:e–1]

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271. Evaluate $\int_0^1 \frac{dx}{1+x^2}$

Sol: Let $I=\int_0^1 \frac{dx}{1+x^2}=\left[\tan^{-1}x\right]_0^1=\tan^{-1}(1)-\tan^{-1}(0)=\frac{\pi}{4}$

---

272. Evaluate $\int_2^3 \frac{xdx}{x^2+1}$

Sol: Let $I=\int_2^3 \frac{x}{x^2+1}dx=\frac{1}{2}\int_2^3 \frac{2x}{x^2+1}dx=\frac{1}{2}\left[\log(1+x^2)\right]_2^3$

$=\frac{1}{2}[\log(1+3^2)-\log(1+2^2)]=\frac{1}{2}[\log(10)-\log(5)]=\frac{1}{2}\log\left(\frac{10}{5}\right)=\frac{1}{2}\log 2$

PQ Evaluate $\int_0^1 \frac{x}{x^2+1}dx$ $\left[\text{Ans}:\frac{1}{2}\log 2\right]$

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273. Evaluate $\int_2^5 (x+2)^2\,dx$

Sol: $I=\int_2^5 (x+2)^2\,dx=\left[\frac{(x+2)^3}{3}\right]_2^5=\frac{1}{3}[(x+2)^3]_2^5=\frac{1}{3}[(5+2)^3-(2+2)^3]=\frac{1}{3}[7^3-4^3]=93$

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274. Find $\int_0^{\pi/2} \sin^2 x\cos x\,dx$ [DI by Substitution]

Sol: Put $\sin x=t\Rightarrow \cos x\,dx=dt$; $x=0\Rightarrow t=\sin 0=0$; $x=\frac{\pi}{2}\Rightarrow t=\sin\frac{\pi}{2}=1$

$\therefore \int_0^{\pi/2} \sin^2 x\cos x\,dx=\int_0^1 t^2 dt=\left[\frac{t^3}{3}\right]_0^1=\frac{1}{3}[1-0]=\frac{1}{3}$

275. Evaluate $\int_0^4 |x-1|\,dx$

Sol: $|x-1| = (x-1) \le 0$ when $0 \le x \le 1$ and $(x-1) \ge 0$ when $1 \le x \le 4$

$I = \int_0^4 |x-1|\,dx = \int_0^1 -(x-1)dx + \int_1^4 (x-1)dx \quad \left(\int_a^b f(x)dx = \int_b^c f(x)dx + \int_c^b f(x)dx\right)$

$= \left[x - \frac{x^2}{2}\right]_0^1 + \left[\frac{x^2}{2} - x\right]_1^4 = \left(1 - \frac{1}{2} - 0\right) + \left[(8-4) - \left(\frac{1}{2} - 1\right)\right] = \frac{1}{2} + \frac{9}{2} = 5$

[Ans:9]

PQ Evaluate $\int_2^8 |x-5|\,dx$.

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276. Evaluate $\int_0^4 \frac{x^2}{1+x}\,dx$

Sol: $I = \int_0^4 \frac{x^2}{1+x}dx = \int_0^4 \frac{(x^2-1)+1}{1+x}\,dx = \int_0^4 \left(\frac{(x-1)\cancel{(x+1)}}{\cancel{1+x}} + \frac{1}{1+x}\right)dx = \int_0^4 (x-1)dx + \int_0^4 \frac{dx}{1+x}$

$= \left[\frac{x^2}{2} - x\right]_0^4 + [\log(1+x)]_0^4 = \frac{4^2}{2} - 4 + \log 5 - \log 1 - 0 = 4 + \log 5$

---

277. Evaluate $\int_0^{\frac{\pi}{2}} \cos^2 x\,dx$

Sol: Let $I = \int_0^{\frac{\pi}{2}} \cos^2 x\,dx = \int_0^{\pi/2} \left(\frac{1+\cos 2x}{2}\right)dx = \left[\frac{x}{2} + \frac{\sin 2x}{4}\right]_0^{\pi/2} = \frac{1}{2}\left[x + \frac{\sin 2x}{2}\right]_0^{\pi/2}$

$= \frac{1}{2}\left[\left(\frac{\pi}{2} + \frac{\sin \pi}{2}\right) - \left(0 + \frac{\sin 0}{2}\right)\right] = \frac{1}{2}\left[\frac{\pi}{2} + 0 - 0 - 0\right] = \frac{\pi}{4}$

---

278. Evaluate $\int_0^{\frac{\pi}{2}} \frac{\sin x}{1+\cos^2 x}dx$

Sol: Put $\cos x = t \Rightarrow -\sin x dx = dt$. When $x = 0$ we have $t = 1$ and when $x = \frac{\pi}{2}$ we have $t = 0$

$\Rightarrow \int_0^{\frac{\pi}{2}} \frac{\sin x}{1+\cos^2 x}dx = -\int_1^0 \frac{dt}{1+t^2} = -\left[\tan^{-1} t\right]_1^0 = -\left[\tan^{-1} 0 - \tan^{-1} 1\right] = -\left[-\frac{\pi}{4}\right] = \frac{\pi}{4}$

---

279. Evaluate $\int_0^{\frac{\pi}{4}} \sin 2x dx$

Sol: $\int_0^{\frac{\pi}{4}} \sin 2x dx = \left[\frac{-\cos 2x}{2}\right]_0^{\frac{\pi}{4}} = \left[\frac{-\cos 2\frac{\pi}{4}}{2}\right] - \left[\frac{-\cos 0}{2}\right] = 0 - \left(\frac{-1}{2}\right) = 0 + \frac{1}{2} = \frac{1}{2}. \quad \left[\because \cos\frac{\pi}{2} = 0\right]$

2 Marks

10 DIFFERENTIAL EQUATIONS

THIS FIRST PAGE IS BEST & PROBLEMS IN 3RD PAGE ARE EASY

280. **Find the order and degree, if** $xy\frac{d^2y}{dx^2}+x\left(\frac{dy}{dx}\right)^2-y\frac{dy}{dx}=0$

Sol: 'Highest order derivative' in the given D.E is $\frac{d^2y}{dx^2}$. ∴ Order of DE is 2.

Power of $\frac{d^2y}{dx^2}$ is 1. ∴ Degree of DE is 1.

PQ **Find the order and degree of** $\frac{d^2y}{dx^2}+5x\left(\frac{dy}{dx}\right)^2-6y=\log x$ **[A:** order is 2, degree is 1.]

---

281. **Determine order and degree of** $\left(\frac{ds}{dt}\right)^4+3s\frac{d^2s}{dt^2}=0$ **[BMP]**

Sol: 'Highest order derivative' in the given DE is $\frac{d^2s}{dt^2}$. ∴ Order of DE is 2.

Power of $\frac{d^2s}{dt^2}$ is 1. ∴ Degree of DE is 1.

PQ **Find the order and degree of** $\left(\frac{dy}{dx}\right)^3-4\left(\frac{dy}{dx}\right)^2+7y=\sin x$ **[A:** Order is 1. Degree is3.]

---

282. **Determine order and degree of** $y''+(y')^2+2y=0$

Sol: 'Highest order derivative' present in the given DE is y". ∴ Order of DE is 2.

Power of y" is one. ∴ Degree of DE is 1.

PQ **Determine order and degree of** $y'''+2y''+y'=0$ [Ans: Order is 3. Degree is 1.]

---

283. **Determine order and degree of** $(y''')^2+(y'')^3+(y')^4+y^5=0$

Sol: 'Highest order derivative' in the given D.E is y'''. ∴ Order of DE is 3.

Power of y''' is 2. ∴ Degree of DE is 1.

PQ **Determine order and degree of** $y''+2y'+\sin y=0$ **[Ans:** Order is two.Degree is 1]

284. **Verify that y=Ax is a solution to** $xy' = y,\ (x \neq 0)$

Sol: Given function is $y = Ax \Rightarrow y' = \frac{d}{dx}(Ax) = A$

Verification: L.H.S=$xy' = xA = Ax = y$ = R.H.S . Hence verified.

---

285. **Verify that the function y = a cos x + b sin x, where, a, b ∈ R is a solution of the differential equation** $\frac{d^2y}{dx^2} + y = 0$

Sol: The given function is $y = a\cos x + b\sin x$ ... (1) $\Rightarrow \frac{dy}{dx} = -a\sin x + b\cos x$

$\frac{d^2y}{dx^2} = -a\cos x - b\sin x = -(a\cos x + b\sin x) = -y \Rightarrow \frac{d^2y}{dx^2} + y = 0$. Hence verified.

---

286. **Verify that y= x^2+2x+C is a solution to** $y' - 2x - 2 = 0$

Sol: Given function is $y = x^2 + 2x + c \Rightarrow y' = \frac{d}{dx}(x^2 + 2x + c) \Rightarrow y' = 2x + 2$

Verification : $y' - 2x - 2 = (2x + 2) - 2x - 2 = 0$. Hence verified.

---

287. **Verify that y=cosx+C is a solution to** $y' + \sin x = 0$

Sol: Given function is $y = \cos x + C \Rightarrow y' = \frac{d}{dx}(\cos x + C) \Rightarrow y' = -\sin x$

Verification: $y' + \sin x = -\sin x + \sin x = 0$. Hence verified.

---

288. **Verify that** $y = \sqrt{1+x^2}$ **is a solution to** $y' = \frac{xy}{1+x^2}$

Sol: Given function is $y = \sqrt{1+x^2} \Rightarrow y' = \frac{d}{dx}\left(\sqrt{1+x^2}\right) = \frac{1}{2\sqrt{1+x^2}} \cdot \frac{d}{dx}(1+x^2)$

$= \frac{2x}{2\sqrt{1+x^2}} = \frac{x}{2\sqrt{1+x^2}} = \frac{x(\sqrt{1+x^2})}{(1+x^2)} = \frac{xy}{1+x^2}$

$\therefore y' = \frac{xy}{1+x^2}$ Hence satisfied and verified.

289. **Find the general solution of the differential equation** $\frac{dy}{dx}=\frac{1+y^2}{1+x^2}$

Sol: Given De is written as $\frac{dy}{1+y^2}=\frac{dx}{1+x^2}\Rightarrow\int\frac{dy}{1+y^2}=\int\frac{dx}{1+x^2}$

$\Rightarrow \tan^{-1}y=\tan^{-1}x+C$ .This is the general solution of given DE.

PQ **Find the general solution of** $\frac{dy}{dx}=(1+x^2)(1+y^2)$ $\left[\text{Ans}: x+\frac{x^3}{x}+C\right]$

---

290. **Find the general solution of the differential equation** $\frac{dy}{dx}+\sqrt{\frac{1-y^2}{1-x^2}}=0$

Sol: Given D.E is $\frac{dy}{dx}+\sqrt{\frac{1-y^2}{1-x^2}}=0\Rightarrow\frac{dy}{dx}=-\frac{\sqrt{1-y^2}}{\sqrt{1-x^2}}\Rightarrow\frac{dy}{\sqrt{1-y^2}}=-\frac{dx}{\sqrt{1-x^2}}$

Integrating both sides, we get $\sin^{-1}y=-\sin^{-1}x+C\Rightarrow\sin^{-1}x+\sin^{-1}y=C$

---

291. **Find the general solution of** $\frac{dy}{dx}+\frac{y}{x}=x^2$

Sol: Given D.E is in the form $\frac{dy}{dx}+Py=Q$. Here $P=\frac{1}{x}$ and $Q=x^2$

$IF=e^{\int Pdx}=e^{\int\frac{1}{x}dx}=e^{\log x}=x$

General Solution is $y(I.F)=\int(Q\times IF)dx+C$

$\therefore yx=\int(x^2.x)dx+C\Rightarrow yx=\int x^3dx+C\Rightarrow xy=\frac{x^4}{4}+C$

---

292. **Find a particular solution of** $\frac{dy}{dx}=y\tan x; y=1$ **when** $x=0$

Sol: Given D.E is $\frac{dy}{dx}=y\tan x\Rightarrow\frac{dy}{y}=\tan x dx$

Integrating both sides, we get $\int\frac{dy}{y}=\int\tan x dx\Rightarrow\log y=\log(\sec x)+\log C$

$\Rightarrow\log y=\log(\sec xC)\Rightarrow y=C\sec x$

Now, $y=1, x=0\Rightarrow 1=C\times\sec 0\Rightarrow 1=C\times 1\Rightarrow C=1$, Thus y=secx is a particular solution.

VSAQ STAR QUESTIONS PLUS

PQ QR Code

SQP QR Code

MATRICES & DETERMINANTS

1. Solve the equation for x, y, z and t, if $2\begin{bmatrix} x & z \\ y & t \end{bmatrix}+3\begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}=3\begin{bmatrix} 3 & 5 \\ 4 & 6 \end{bmatrix}$

2. Given , $3\begin{bmatrix} x & y \\ z & w \end{bmatrix}=\begin{bmatrix} x & 6 \\ -1 & 2w \end{bmatrix}+\begin{bmatrix} 4 & x+y \\ z+w & 3 \end{bmatrix}$ find the values of x, y, z and w.

3. Simplify $\cos\theta\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}+\sin\theta\begin{bmatrix} \sin\theta & -\cos\theta \\ \cos\theta & \sin\theta \end{bmatrix}$

4. For the matrix $A=\begin{bmatrix} 1 & 5 \\ 6 & 7 \end{bmatrix}$, verify that $(A+A')$ is a symmetric matrix

5. For the matrix $A=\begin{bmatrix} 1 & 5 \\ 6 & 7 \end{bmatrix}$, verify that $(A-A')$ is a skew symmetric matrix

6. Find the minor of element 6 in the determinant $\Delta=\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}$

7. If $A'=\begin{bmatrix} -2 & 3 \\ 1 & 2 \end{bmatrix}$ and $B=\begin{bmatrix} -1 & 0 \\ 1 & 2 \end{bmatrix}$, then find $(A+2B)'$

CONTINUITY & DIFFERENTIABILITY

8. Find $\frac{dy}{dx}$ if $x-y=\pi$.

9. Find $\frac{dy}{dx}$, if $y+\sin y=\cos x$

10. Find $\frac{dy}{dx}$ of $2x+3y=\sin x$

11. Find $\frac{dy}{dx}$ of $2x+3y=\sin y$

12. Find $\frac{dy}{dx}$ of $\sin^2 x+\cos^2 y=1$

13. Find $\frac{dy}{dx}$ of $y=\sec^{-1}\left(\frac{1}{2x^2-1}\right), 0<x<\frac{1}{\sqrt{2}}$

14. Differentiatelog $(\cos e^x)$ w.r.t. x.

15. Differentiate $\cos(\log x + e^x)$, $x > 0$ w.r.t. x.

INTEGRALS

INDEFINITE INTEGRALS & DEFINITE INTEGRALS

16. Find integral of $\frac{1}{x + x\log x}$

17. Find integral of $\sqrt{ax+b}$

18. Find integral of e^{2x+3}

19. Find integral of $\frac{e^{2x}-1}{e^{2x}+1}$

20. Find integral of $\sqrt{\sin 2x}\cos 2x$

21. Find integral of $\frac{\sin x}{1+\cos x}$

22. Find $\int \frac{dx}{x^2-6x+13}$

23. Find the integral of $\frac{x^2}{1-x^6}$

24. Evaluate $\int_2^3 \frac{1}{x} dx$

25. Evaluate $\int_1^2 (4x^3 - 5x^2 + 6x + 9) dx$

26. Evaluate $\int_0^{\frac{\pi}{4}} \tan x dx$

27. Evaluate $\int_0^1 \frac{dx}{\sqrt{1-x^2}}$

28. Evaluate $\int_2^3 \frac{dx}{x^2-1}$

29. Evaluate $\int_0^{\pi/2} \sin^3 x \, dx$